

Foundations of DenseNet

[DenseNet]

1.0 Content Block

$$\begin{aligned} x_{\ell+2} &= F(x_{\ell+1}) + x_{\ell+1} = F(x_{\ell+1}) + F(x_{\ell}) + x_{\ell} \\ &= F(x_{\ell+1}) + x_{\ell+1} = F(x_{\ell+1}) + F(x_{\ell}) + x_{\ell} \end{aligned}$$

2.0 Content Block

Previous work Residual connections were noticed in neuroanatomy, such as Lorente de No (1938). McCulloch and Pitts (1943) proposed artificial neural networks and considered those with residual connections. In 1961, Frank Rosenblatt described a three-layer multilayer perceptron (MLP) model with skip connections. The model was referred to as a "cross-coupled system", and the skip connections were forms of cross-coupled connections.

3.0 Content Block

Backward propagation The residual learning formulation provides the added benefit of mitigating the vanishing gradient problem to some extent. However, it is crucial to acknowledge that the vanishing gradient issue is not the root cause of the degradation problem, which is tackled through the use of normalization. To observe the effect of residual blocks on backpropagation, consider the partial derivative of a loss function E with respect to some residual block input x_{ℓ} . Using the equation above from forward propagation for a later residual block $L > \ell$:

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$$\frac{\partial E}{\partial x} = \frac{\partial E}{\partial x} \frac{\partial x}{\partial x} = \frac{\partial E}{\partial x} (1 + \frac{\partial x}{\partial x} \sum_{i=1}^{L-1} F(x_i)) = \frac{\partial E}{\partial x} + \frac{\partial E}{\partial x} \frac{\partial x}{\partial x} \sum_{i=1}^{L-1} F(x_i)$$

5.0 Content Block

Degradation problem Sepp Hochreiter discovered the vanishing gradient problem in 1991 and argued that it explained why the then-prevalent forms of recurrent neural networks did not work for long sequences. He and Schmidhuber later designed the LSTM architecture to solve this problem, which has a "cell state" c_t that can function as a generalized residual connection. The highway network (2015) applied the idea of an LSTM unfolded in time to feedforward neural networks, resulting in the highway network. ResNet is equivalent to an open-gated highway network.

6.0 Content Block

Basic block A basic block is the simplest building block studied in the original ResNet. This block consists of two sequential 3x3 convolutional layers and a residual connection. The input and output dimensions of both layers are equal.

7.0 Content Block

Projection connection If the function F is of type $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ where $n \neq m$, then $F(x) + x$ is

undefined. To handle this special case, a projection connection is used:

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A residual neural network (also referred to as a residual network or ResNet) is a deep learning architecture in which the layers learn residual functions with reference to the layer inputs. It was developed in 2015 for image recognition, and won the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) of that year. As a point of terminology, "residual connection" refers to the specific architectural motif of $x + f(x)$, where f is an arbitrary neural network module. The motif had been used previously (see §History for details). However, the publication of ResNet made it widely popular for feedforward networks, appearing in neural networks that are seemingly unrelated to ResNet. The residual connection stabilizes the training and convergence of deep neural networks with hundreds of layers, and is a common motif in deep neural networks, such as transformer models (e.g., BERT, and GPT models such as ChatGPT), the AlphaGo Zero system, the AlphaStar system, and the AlphaFold system.

9.0 Content Block

Forward propagation If the output of the ℓ -th residual block is the input to the $(\ell + 1)$ -th residual block (assuming no activation function between blocks), then the $(\ell + 1)$ -th input is:

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During the early days of deep learning, there were attempts to train increasingly deep models. Notable examples included the AlexNet (2012), which had 8 layers, and the VGG-19 (2014), which had 19 layers. However, stacking too many layers led to a steep reduction in training accuracy, known as the "degradation" problem. In theory, adding additional layers to deepen a network should not result in a higher training loss, but this is what happened with VGGNet. If the extra layers can be set as identity mappings, however, then the deeper network would represent the same function as its shallower counterpart. There is some evidence that the optimizer is not able to approach identity mappings for the parameterized layers, and the benefit of residual connections was to allow identity mappings by default. In 2014, the state of the art was training deep neural networks with 20 to 30 layers. The research team for ResNet attempted to train deeper ones by empirically testing various methods for training deeper networks, until they came upon the ResNet architecture.

11.0 Content Block

where L is the index of a residual block and ℓ is the index of some earlier block. This formulation suggests that there is always a signal that is directly sent from a shallower block ℓ to a deeper block L .

12.0 Content Block

where P is typically a linear projection, defined by $P(x) = Mx$ where M is a $m \times n$ matrix. The matrix is trained via backpropagation, as is any other parameter of the model.

13.0 Content Block

Applications Originally, ResNet was designed for computer vision. All transformer architectures include residual connections. Indeed, very deep transformers cannot be trained without them. The original ResNet paper made no claim on being inspired by biological systems. However, later research has related ResNet to biologically-plausible algorithms. A study published in Science in 2023 disclosed the complete connectome of an insect brain (specifically that of a fruit fly larva). This study discovered "multilayer shortcuts" that resemble the skip connections in artificial neural networks, including ResNets.